Radiation-Free Spinal Surface Scanner

Pipeline, methods and results report

Fringe-projection back-surface scanning for spinal deformity assessment

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Scan Code7701

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1. Summary

Four R scripts (**refcal**, **surfcolour**, **blob**, **clinic2**) turn a single photograph of a patient's bare back into a 3D surface map, a set of spinal landmarks and a clinical deformity report, with no X-ray exposure. A horizontal grating (fringe) pattern is projected onto the back, and the way the fringes bend encodes the surface height. Coloured marker stickers on the spine and pelvis supply the anatomical reference frame.

For scan **Code7701** the pipeline reports a back length of 386 mm, a trunk rotation of -1°, a coronal imbalance of 2 mm, a surface kyphosis angle of 52° and a lordosis angle of 44°, with small left/right asymmetries (3° right and 4° left lateral; 2 L and 4 R volumetric).

Basis of this report: the written description of the scripts, the pipeline flowchart, and the output files results.pdf and results.txt. The R source itself was not provided, so the stage descriptions stay general and do not state exact formulas used inside the scripts.

2. Background and purpose

Spinal deformities such as scoliosis, and abnormal kyphosis or lordosis, are normally measured on X-ray images using the Cobb angle. Because follow-up often needs repeated imaging, there is interest in surface-based methods that can be repeated as often as needed without ionising radiation. This project follows that idea: the shape of the back surface and the position of skin markers over bony landmarks are measured optically, and clinically meaningful parameters are computed from them.

The goals of the code are therefore to (1) reconstruct the back surface in millimetres from one photograph, (2) locate the anatomical landmarks automatically with a manual override, (3) put every scan into the same standard orientation, and (4) report the deformity measures in a form a clinician can read at a glance.

3. Principle of operation: fringe projection

A projector casts a pattern of parallel horizontal stripes onto the back. On a flat reference plane the stripes are straight and evenly spaced. On a curved surface such as a back, the stripes are displaced up or down in proportion to the local height. The camera records this distorted pattern, and the displacement is extracted using **Fourier Transform Profilometry (FTP)**.

1. **Fourier transform** of the image: the fringe pattern shows up as a strong peak at the fringe frequency, separate from the slowly varying background.
2. **Filtering** keeps only that peak, and the **inverse transform** gives a complex signal whose angle is the fringe phase.
3. **Phase difference** between the patient image and the flat reference plane is proportional to height.
4. **Phase unwrapping** removes the 2pi jumps that arise because phase is only known modulo one cycle.
5. **Scaling** with the calibration constants (camera/projector geometry and fringe frequency) converts the unwrapped phase difference to millimetres.

In the standard textbook form for this geometry, the height is related to the phase difference by:

```
h(x,y) = L * dphi(x,y) / ( dphi(x,y) - 2*pi*f0*d )
```

where L is the distance from the camera to the reference plane, d is the separation between camera and projector axes, f0 is the fringe frequency on the reference plane, and dphi is the unwrapped phase difference. This is the general FTP relation, quoted to explain the method; the exact expression coded in the scripts may differ in form. This is why calibration (the first script) must provide the geometry and the fringe frequency.

4. Pipeline overview

The flowchart shows how data moves between the scripts. Each coloured box is an R script; file icons are the intermediate and final files passed between them. Calibration feeds the surface script, whose marker output can be corrected by hand before the clinical script produces the report files.

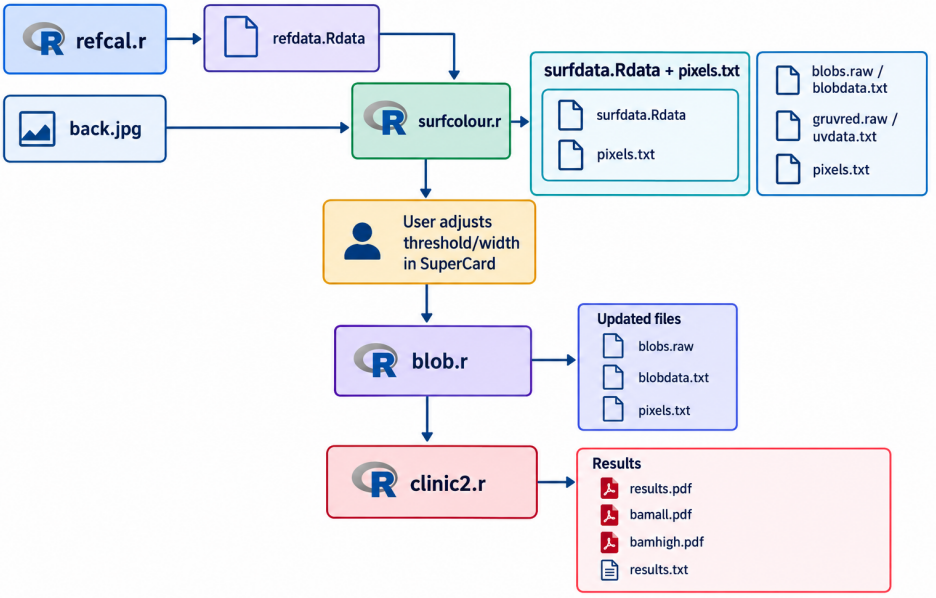


Figure 1. Pipeline flowchart: refcal.r, surfcolour.r, a manual SuperCard threshold step, blob.r and clinic2.r.

Table 1. The four scripts at a glance

Script	Inputs	Main operations	Outputs
refcal.r	Reference-plane photograph	Computes camera/projector geometry and fringe frequency; run once per rig setup	refdata.Rdata
surfcolour.r	back.jpg, refdata.Rdata	Autocrop, arm removal, FFT demodulation, phase unwrapping, height conversion; marker detection	surfdata.Rdata, pixels.txt, blobs.raw, blobdata.txt, gruvred.raw, uvdata.txt
blob.r	UV image and user threshold/width settings	Re-thresholds the UV image to refine marker detection	Updated blobs.raw, blobdata.txt, pixels.txt
clinic2.r	Surface and marker files	Re-orients surface, fits 5th-order spine curve, computes clinical parameters, draws report	results.pdf, bamall.pdf, bamhigh.pdf, results.txt

5. Stage-by-stage: what each script does to the data

5.1 Calibration (refcal.r; called refcalvert.r in the written description)

- **Run once per rig setup**, using a photograph of a flat reference plane.
- Computes the **camera/projector geometry** and the **fringe frequency** of the projected grating.
- Saves these constants to `refdata.Rdata`, reused by every later scan. If the camera, projector or their distance changes, calibration must be repeated, because the height scale depends on it.

5.2 Surface reconstruction and marker detection (surfcolour.r)

Inputs are the patient photograph (`back.jpg`) and `refdata.Rdata`. The script does two jobs at the same time.

A. Building the 3D surface

1. **Autocrop** the photo to the back region.
2. **Remove the arms** by detecting the armpits, so they do not distort the trunk surface.
3. **Demodulate the fringe pattern** with the Fourier transform to obtain the phase map.
4. **Unwrap the phase**, so it is continuous across the back.
5. **Convert phase to height** using the calibration constants, giving the height map in mm.

B. Finding the anatomical markers

Coloured stickers on the spine and the PSIS (the bony points at the back of the pelvis) are located in the image. They define the reference frame: the PSIS line gives the horizontal, and the spinal stickers give the path of the spine.

Outputs: `surfdata.Rdata` and `pixels.txt` (surface and pixel data), and `blobs.raw / blobdata.txt` and `gruvred.raw / uvdata.txt` (the detected marker blobs and the UV image used to find them).

5.3 Manual fine-tuning (SuperCard) and blob.r

- If automatic marker detection is imperfect (a missed sticker, or a false detection on skin or background), the user adjusts the **threshold and width** settings in SuperCard.
- `blob.r` then **re-thresholds the UV image** with those settings and rewrites blobs.raw, blobdata.txt and pixels.txt.
- This is the only human-in-the-loop step. It matters because the markers decide the whole coordinate frame, so a wrong marker would give wrong angles downstream.

5.4 Clinical analysis and report (clinic2.r)

1. **Load** the reconstructed surface and markers.
2. **Rotate and tilt** the surface so the PSIS line is horizontal and the vertebra prominens (C7) lies directly above the sacrum. After this, all measurements share one standard orientation.
3. **Fit a 5th-order polynomial** through the spinal markers to get a smooth spine curve in the coronal (front-on) and sagittal (side-on) planes.
4. **Extract clinical parameters**: Cobb-like angles of the lateral curve, coronal imbalance, axial trunk rotation, transverse skin angles, kyphosis and lordosis angles, scapular height difference, and bilateral and volumetric asymmetry.
5. **Render outputs**: the multi-page report results.pdf, two asymmetry maps (bamall.pdf, bamhigh.pdf) and the numeric summary results.txt.

6. Files and data flow

The scripts communicate only through files, which makes each stage easy to re-run on its own. For example, if only the marker threshold needs changing, blob.r and clinic2.r can be re-run without repeating the surface reconstruction.

Table 2. Files in the pipeline

File	Created by	Used by	Content
refdata.Rdata	refcal.r	surfcolour.r	Calibration constants: geometry and fringe frequency
back.jpg	Camera	surfcolour.r	Photograph of the back with fringes projected
surfdata.Rdata	surfcolour.r	clinic2.r	Reconstructed surface data
pixels.txt	surfcolour.r, blob.r	clinic2.r	Pixel data linked to the surface and markers
blobs.raw, blobdata.txt	surfcolour.r, blob.r	clinic2.r	Detected marker blobs and their data
gruvred.raw, uvdata.txt	surfcolour.r	blob.r	UV image and data used for marker thresholding
results.pdf	clinic2.r	Clinician	Multi-page report: height map, contour plot, three views
bamall.pdf, bamhigh.pdf	clinic2.r	Clinician	Bilateral asymmetry maps
results.txt	clinic2.r	Further analysis	Numeric summary (unlabelled column of values)

7. Results for scan Code7701

Figure 2 is the report page produced by clinic2.r (results.pdf). Top row: the reconstructed height map and the contour plot (heights in mm; blue dots are the spinal and pelvic markers). Bottom row: transverse skin-angle profiles, the coronal view and the sagittal view.

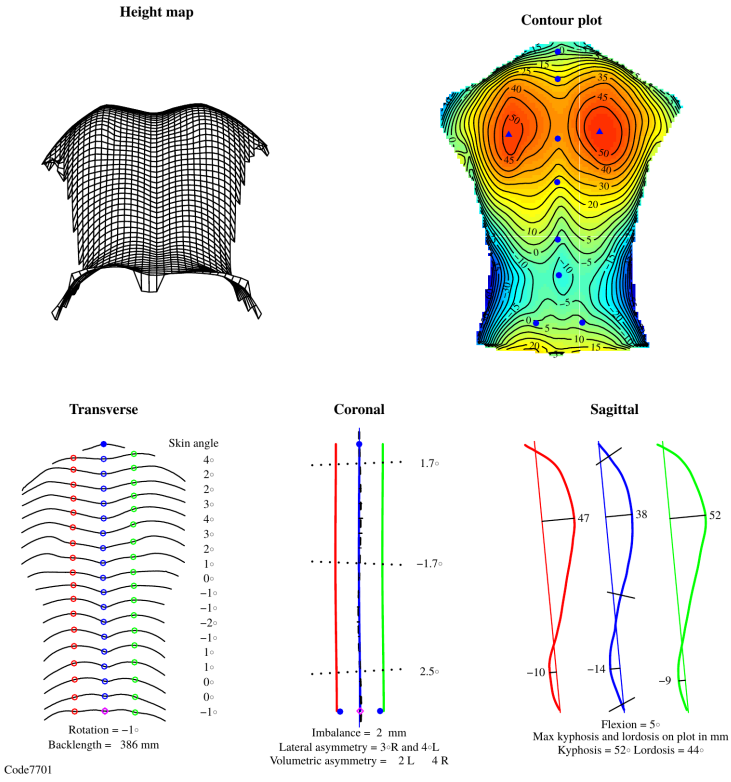


Figure 2. Output of clinic2.r for scan Code7701 (results.pdf).

7.1 Reading the panels

- **Height map and contour plot:** the wire-frame and colour map show the back surface from about -60 mm to +50 mm. Two raised regions on either side of the upper spine correspond to the shoulder-blade areas, and the spine itself runs down the central valley.
- **Transverse:** each curve is a horizontal slice of the back at one level, with the marker positions on it. The number beside each slice is the skin angle at that level. Rotation (-1°) and back length (386 mm) are printed underneath.
- **Coronal:** the front-on view of the fitted spine curve with imbalance (2 mm), lateral asymmetry (3° R and 4° L) and volumetric asymmetry (2 L, 4 R). Skin angles at three levels are 1.7°, -1.7° and 2.5°.
- **Sagittal:** the side-on view for three profiles (red, blue and green), with the maximum kyphosis and lordosis offsets in mm marked on each. Flexion is 5°, kyphosis 52° and lordosis 44°.

Table 3. Values read from results.pdf

Measure	Result	Where it appears
Back length	386 mm	Transverse panel
Trunk rotation	-1°	Transverse panel
Coronal imbalance	2 mm	Coronal panel
Flexion	5°	Sagittal panel
Kyphosis	52°	Sagittal panel (fitted spine curve)
Lordosis	44°	Sagittal panel (fitted spine curve)
Lateral asymmetry	3° right and 4° left	Coronal panel
Volumetric asymmetry	2 (left) and 4 (right)	Coronal panel
Coronal skin angles	1.7°, -1.7°, 2.5°	Coronal panel, three levels
Transverse skin angles	-2° to 4° across 18 levels	Transverse panel
Max kyphosis / lordosis offsets (mm)	47 / -10, 38 / -14, 52 / -9	Sagittal panel, three profiles (red, blue, green)
Height map range	about -60 mm to +50 mm	Height map and contour plot

Note on results.txt. It is an unlabelled column of numbers. Entries such as 386, -1, 5, 2, 52 and 44 line up with the labelled PDF values, but the others (for example -0.07, -0.45, -0.85, -0.57, 69 and 60) have no labels in the file, so they are not interpreted here. A header row in the output would make the file self-describing.

8. What each clinical measure means

Table 4. Clinical parameters produced by clinic2.r

Parameter	Plane	What it describes	Value for Code7701
Cobb-like angle	Coronal	Sideways curvature of the spine, estimated from the fitted curve (an estimate, not the radiographic Cobb angle)	See coronal panel
Coronal imbalance	Coronal	Sideways offset of the upper spine from the line above the sacrum	2 mm
Axial rotation	Transverse	Twist of the trunk about the long axis; a rib hump shows up as rotation	-1°
Transverse skin angle	Transverse	Slope of the skin surface across the back at a given level; left/right differences indicate asymmetry	-2° to 4°
Kyphosis angle	Sagittal	Forward curve of the upper back	52°
Lordosis angle	Sagittal	Inward curve of the lower back	44°
Flexion	Sagittal	Forward lean of the trunk	5°
Scapular height difference	Surface	Height difference between the two shoulder-blade regions	Not printed in this report page
Lateral asymmetry	Coronal	Left versus right surface difference	3° R and 4° L
Volumetric asymmetry	Surface	Left versus right volume difference (also mapped in bamall.pdf and bamhigh.pdf)	2 L, 4 R

9. How to read the results

- **Frontal plane (coronal):** a 2 mm imbalance and skin angles of only a few degrees mean the trunk is close to vertical and close to symmetric. The 3° right and 4° left lateral asymmetry values are small and similar in size, so no strong one-sided pattern shows up.
- **Rotation:** -1° is a minor axial rotation. In scoliosis screening rotation is a key sign, because a rib hump appears as surface asymmetry; here it is small.
- **Side view (sagittal):** a kyphosis of 52° and a lordosis of 44° describe the thoracic and lumbar curves. Surface-derived angles come from the skin and the fitted marker curve, not from the vertebral endplates, so

they should not be compared directly with radiographic Cobb angles. Against commonly quoted radiographic thoracic ranges, 52° sits at the upper end of normal or mildly increased, which needs a clinician's judgement and ideally a repeat scan or radiographic comparison.

- **Volumetric asymmetry (2 L, 4 R):** the right side carries somewhat more volume than the left, in the same direction as the other small asymmetries but of modest size.
- **Overall:** the scan describes a well-balanced trunk with minimal rotation, small left/right differences and sagittal curves in a plausible range.

10. Suggested operating procedure and quality checks

The sequence below follows the flowchart. The quality checks are suggestions for good practice based on how the pipeline works, not requirements stated in the code.

1. **Set up the rig** and photograph the flat reference plane; run refcal.r. Repeat whenever the camera, projector or distances change.
2. **Place the marker stickers** on the spinal landmarks and the PSIS, then photograph the patient with the fringes projected (back.jpg).
3. **Run surfcolour.r** and check that the back is cropped correctly and the arms are removed.
4. **Check the markers** in the UV image. If any is missing or false, adjust threshold/width in SuperCard and run blob.r.
5. **Run clinic2.r** and review results.pdf, bamall.pdf and bamhigh.pdf.

Table 5. Quality-check list (suggested)

Check	Why it matters
Fringes are sharp and evenly lit across the back	Poor fringe contrast gives noisy phase and a rough height map
Patient stands in the same position at each visit	Posture changes alter the surface and the angles, which hides real change
All spinal and PSIS markers are detected exactly once	Markers define the reference frame; a wrong marker rotates or bends every result
Height map is smooth, without tears or large jumps	Jumps suggest phase-unwrapping errors
Calibration is current for the present rig	A changed rig with an old calibration gives wrong height scale
Results are compared only with scans from the same system	Surface angles are not identical to radiographic angles

11. Limitations to keep in mind

- This is a **screening and monitoring tool**, not a diagnosis. Surface measures estimate, but do not replace, radiographic measures of the bones.
- **Marker placement matters.** The reference frame, the spine curve and almost every angle depend on correct sticker placement and detection, which is why the manual threshold step exists.
- **Calibration must match the setup.** A changed rig without a new calibration run would give wrong heights.
- **Posture and movement** during the photo affect the surface, so repeat scans should use the same standing position.
- A **5th-order polynomial** is smooth and stable but can slightly flatten sharp local curves.
- **Fringe projection** can struggle with steep surface changes, shadows and very uneven lighting, which makes phase unwrapping harder.
- The results file is unlabelled, which makes automated use of the numbers error-prone.

12. Possible improvements

- Add a header row or labelled key-value output to results.txt so the numbers can be used directly in other tools.

- Validate the surface angles against radiographic Cobb angles on a set of patients, to quantify agreement.
- Store the SuperCard threshold and width settings used for each scan so the marker step is reproducible.
- Print the scapular height difference and the full list of parameters on the report page.
- Log repeat-scan differences over time so change in a patient can be tracked automatically.

13. Glossary

Table 6. Terms used in this report

Term	Meaning
Fringe projection	Projecting a stripe pattern on an object and measuring how the stripes bend to find its shape
FTP	Fourier Transform Profilometry: extracting surface height from one fringe image using the Fourier transform
Phase unwrapping	Turning phase that jumps by 2pi into a continuous map
PSIS	Posterior superior iliac spine: bony points at the back of the pelvis used to define the horizontal
Vertebra prominens (C7)	The most prominent vertebra at the base of the neck, used as the top spinal landmark
Coronal plane	The front-on view of the body
Sagittal plane	The side-on view of the body
Transverse plane	The horizontal slice through the body
Kyphosis	Forward curvature of the upper back
Lordosis	Inward curvature of the lower back
Cobb angle	The standard radiographic measure of spinal curve size
Skin angle	Slope of the skin surface at a given level of the back
Volumetric asymmetry	Difference in volume between the left and right halves of the back

14. Conclusion

The code forms a complete chain from one fringe-projection photograph to a clinical report: calibrate the rig once, reconstruct the back surface by Fourier Transform Profilometry, locate the markers, optionally correct detection by hand, then re-orient the data into a standard frame and extract the deformity measures. For scan Code7701 it shows a well-balanced trunk with minimal rotation and small left/right asymmetries, and a kyphosis and lordosis of 52° and 44°. The approach gives repeatable, radiation-free monitoring, with accuracy resting on careful calibration and marker detection.